



Cisco Catalyst 2960-X

Secure Sanitization, IOS Recovery & Engineering Validation

Technical Evidence Report | WS-C2960X-24TS-L | Tera Term serial console | Isolated bench

Scope and evidence standard. This report interprets 12 provided terminal captures. It does not assert unseen tests were completed. The source includes reset/zeroization execution, IOS recovery, boot setup, and configuration/inventory checks. Serial-number fields have been blurred in included screenshots.

Publication warning. Screenshots from company equipment and procedures may be subject to internal restrictions. Written authorization and a privacy/security review are required before posting externally.

Recorded workflow

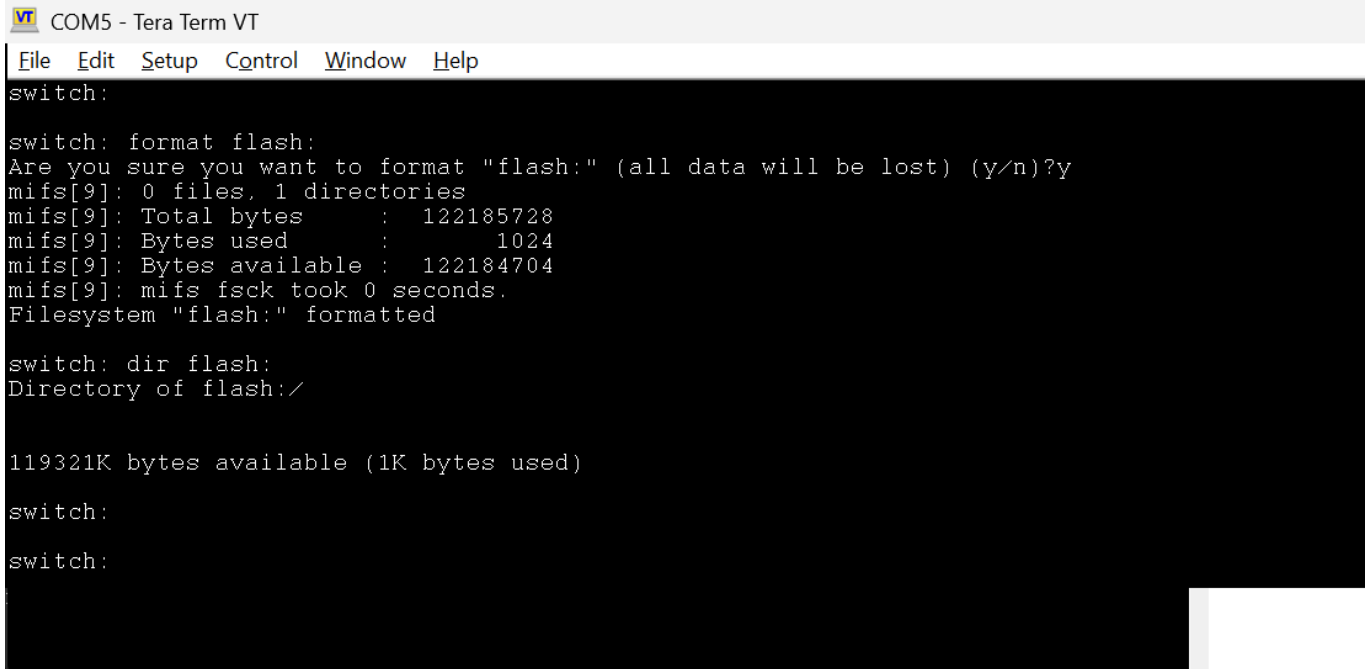
01. Flash formatting and directory check — format flash;; dir flash:
02. Booting approved IOS from USB — boot usbflash0:c2960x-universalk9-mz.152-7.E11.bin
03. Secure factory-reset invocation — factory-reset all secure
04. FIPS zeroization procedure — conf t; fips zeroize
05. Bootloader environment inspection — set; dir usbflash0:
06. USB boot after zeroization — boot usbflash0:c2960x-universalk9-mz.152-7.E11.bin
07. Flash recovery and persistent boot configuration — copy usbflash0: flash;; boot system flash:...; copy run start; show boot
08. Running configuration inspection — show run
09. Startup configuration inspection — show start
10. VLAN database inspection — show vlan
11. VTP and PoE capability check — show vtp status; show power inline
12. Device identity validation — show inv

Evidence walkthrough

01 | Flash formatting and directory check

Commands observed: `format flash;` `dir flash:`

The visible terminal output shows flash formatting and a follow-up directory listing with minimal occupied space. This is one phase of the prescribed removal workflow, not standalone proof of irreversible sanitization.

A terminal window titled "COM5 - Tera Term VT" with a menu bar (File, Edit, Setup, Control, Window, Help). The terminal output shows a user at a "switch:" prompt typing "format flash:". The system responds with a confirmation message: "Are you sure you want to format 'flash:' (all data will be lost) (y/n)?y". This is followed by disk statistics: "mifs[9]: 0 files, 1 directories", "mifs[9]: Total bytes : 122185728", "mifs[9]: Bytes used : 1024", "mifs[9]: Bytes available : 122184704", and "mifs[9]: mifs fsck took 0 seconds." The final message is "Filesystem 'flash:' formatted". The user then types "dir flash:" and the system responds with "Directory of flash:/" and "119321K bytes available (1K bytes used)". The prompt returns to "switch:".

```
COM5 - Tera Term VT
File Edit Setup Control Window Help
switch:
switch: format flash:
Are you sure you want to format "flash:" (all data will be lost) (y/n)?y
mifs[9]: 0 files, 1 directories
mifs[9]: Total bytes      : 122185728
mifs[9]: Bytes used       : 1024
mifs[9]: Bytes available  : 122184704
mifs[9]: mifs fsck took 0 seconds.
Filesystem "flash:" formatted

switch: dir flash:
Directory of flash:/

119321K bytes available (1K bytes used)

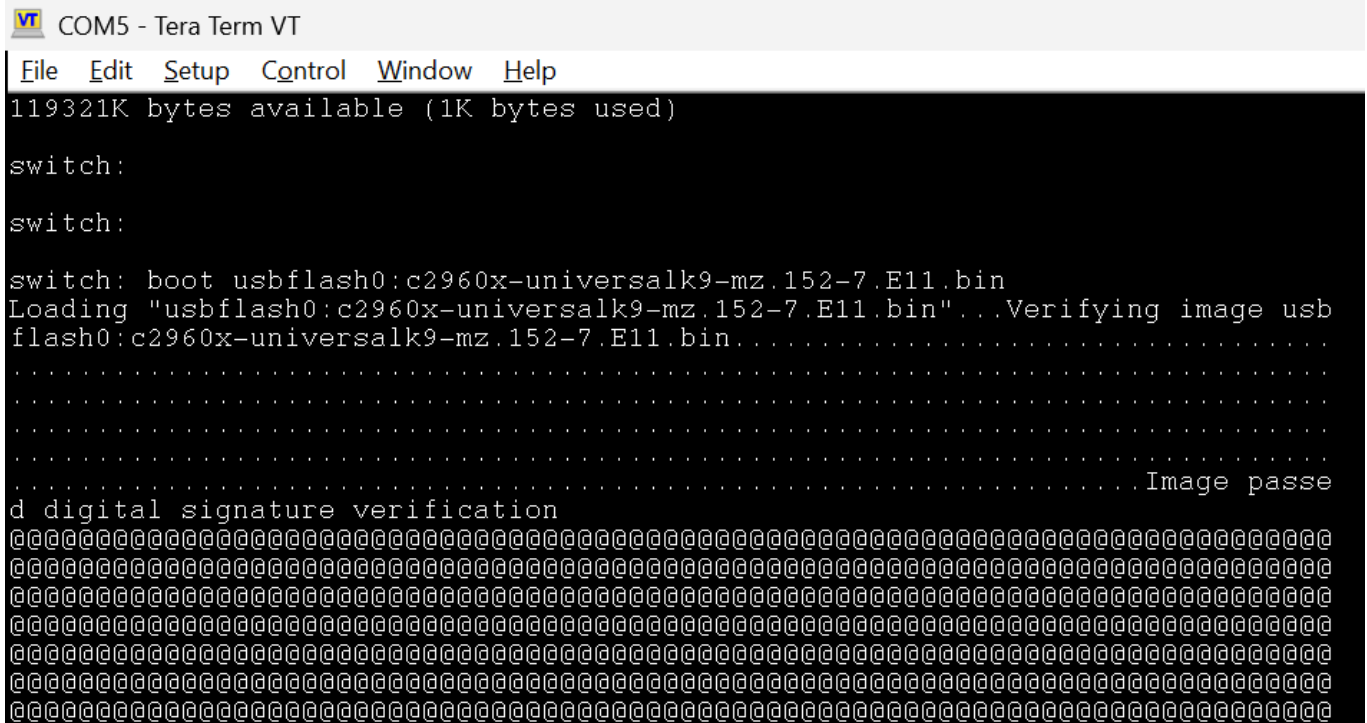
switch:
switch:
```

Figure 1. Redacted original terminal screenshot supplied for this project.

02 | Booting approved IOS from USB

Commands observed: `boot usbflash0:c2960x-universalk9-mz.152-7.E11.bin`

The bootloader loads the selected 2960-X image from removable storage and displays image-signature verification output.



```
COM5 - Tera Term VT
File Edit Setup Control Window Help
119321K bytes available (1K bytes used)

switch:
switch:

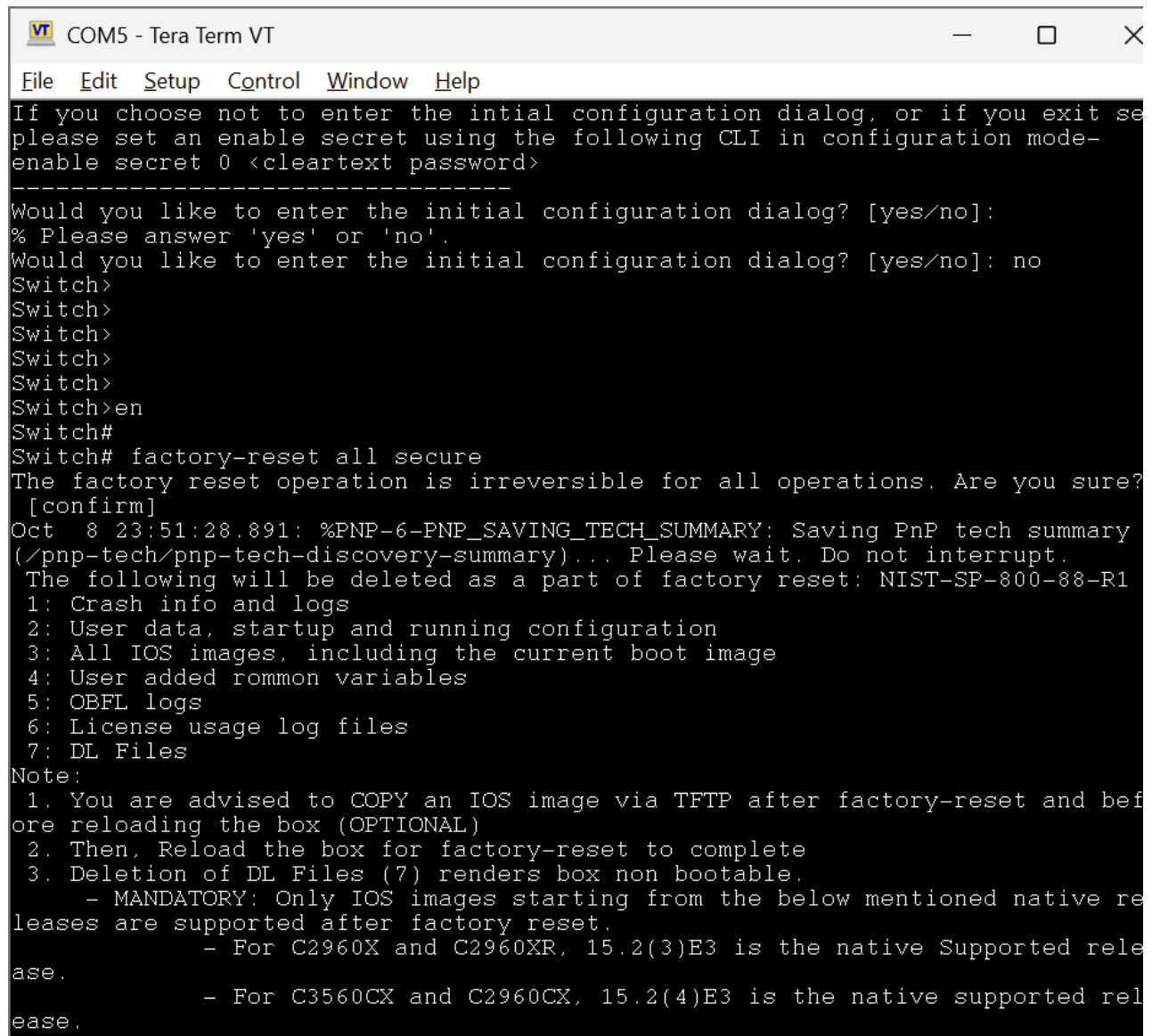
switch: boot usbflash0:c2960x-universalk9-mz.152-7.E11.bin
Loading "usbflash0:c2960x-universalk9-mz.152-7.E11.bin"...Verifying image usb
flash0:c2960x-universalk9-mz.152-7.E11.bin.....
.....
.....Image passed
digital signature verification
eeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeee
eeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeee
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eeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeee
```

Figure 2. Redacted original terminal screenshot supplied for this project.

03 | Secure factory-reset invocation

Commands observed: `factory-reset all secure`

The console displays the destructive reset confirmation and list of categories targeted for removal. The screenshot does not by itself establish the final complete outcome.



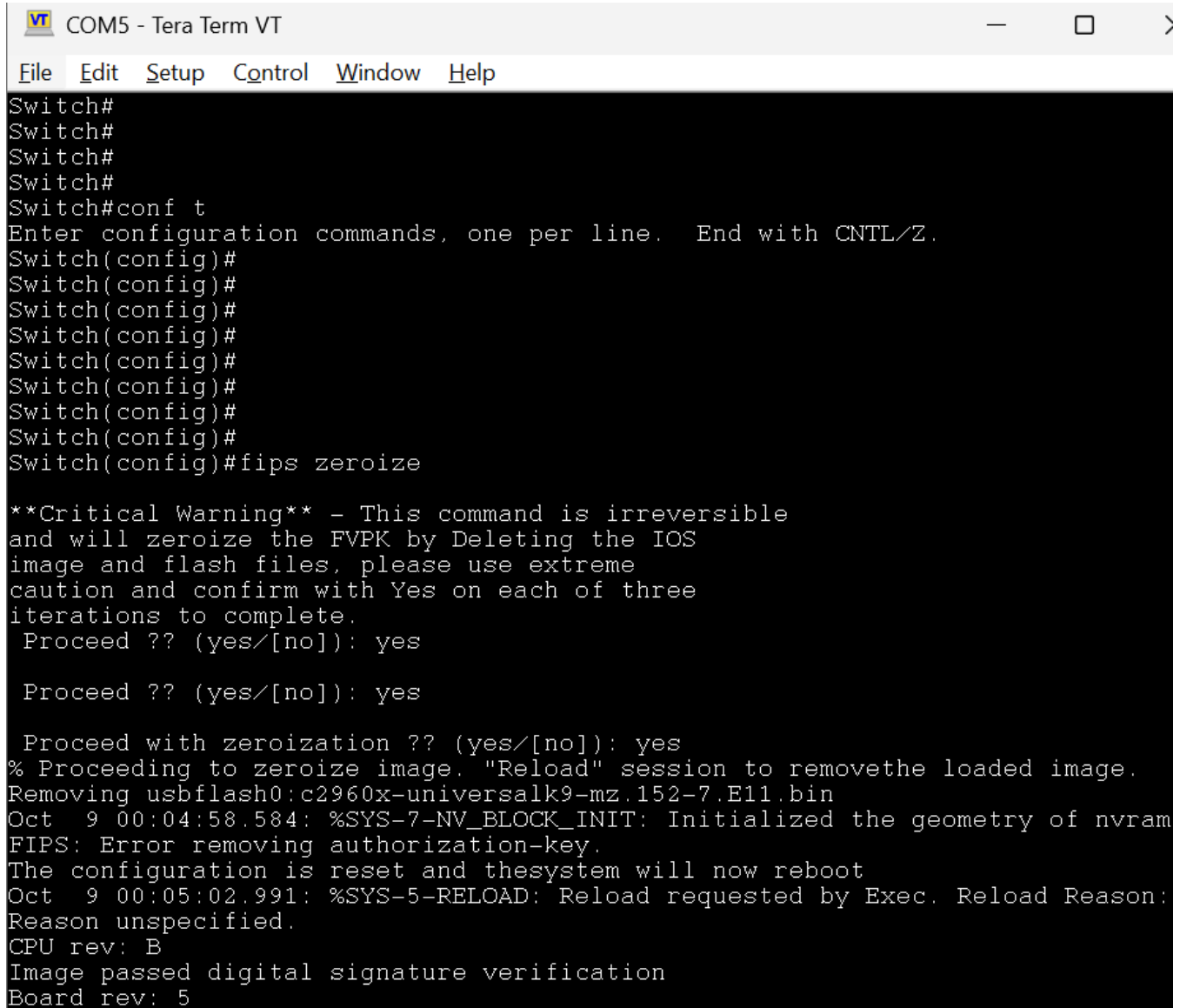
```
COM5 - Tera Term VT
File Edit Setup Control Window Help
If you choose not to enter the initial configuration dialog, or if you exit se
please set an enable secret using the following CLI in configuration mode-
enable secret 0 <cleartext password>
-----
Would you like to enter the initial configuration dialog? [yes/no]:
% Please answer 'yes' or 'no'.
Would you like to enter the initial configuration dialog? [yes/no]: no
Switch>
Switch>
Switch>
Switch>
Switch>
Switch>en
Switch#
Switch# factory-reset all secure
The factory reset operation is irreversible for all operations. Are you sure?
[confirm]
Oct  8 23:51:28.891: %PNP-6-PNP_SAVING_TECH_SUMMARY: Saving PnP tech summary
(/pnp-tech/pnp-tech-discovery-summary)... Please wait. Do not interrupt.
The following will be deleted as a part of factory reset: NIST-SP-800-88-R1
1: Crash info and logs
2: User data, startup and running configuration
3: All IOS images, including the current boot image
4: User added rommon variables
5: OBFL logs
6: License usage log files
7: DL Files
Note:
1. You are advised to COPY an IOS image via TFTP after factory-reset and bef
ore reloading the box (OPTIONAL)
2. Then, Reload the box for factory-reset to complete
3. Deletion of DL Files (7) renders box non bootable.
   - MANDATORY: Only IOS images starting from the below mentioned native re
leases are supported after factory reset.
     - For C2960X and C2960XR, 15.2(3)E3 is the native Supported rele
ase.
     - For C3560CX and C2960CX, 15.2(4)E3 is the native supported rel
ease.
```

Figure 3. Redacted original terminal screenshot supplied for this project.

04 | FIPS zeroization procedure

Commands observed: `conf t; fips zeroize`

The irreversible zeroization prompts and restart-related output are visible. This is part of the controlled sanitization sequence.

A terminal window titled "COM5 - Tera Term VT" with a menu bar (File, Edit, Setup, Control, Window, Help). The terminal output shows a sequence of commands and system responses. It starts with "Switch#" prompts, followed by "conf t" which enters configuration mode. The user enters "fips zeroize", which triggers a series of warnings and confirmation prompts. The system proceeds with zeroizing the image, removing the USB flash, and initializing NV block. It then resets the configuration and reboots the system. The output ends with CPU and board revision information.

```
Switch#
Switch#
Switch#
Switch#
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#
Switch(config)#
Switch(config)#
Switch(config)#
Switch(config)#
Switch(config)#
Switch(config)#fips zeroize

**Critical Warning** - This command is irreversible
and will zeroize the FVPK by Deleting the IOS
image and flash files, please use extreme
caution and confirm with Yes on each of three
iterations to complete.
  Proceed ?? (yes/[no]): yes

  Proceed ?? (yes/[no]): yes

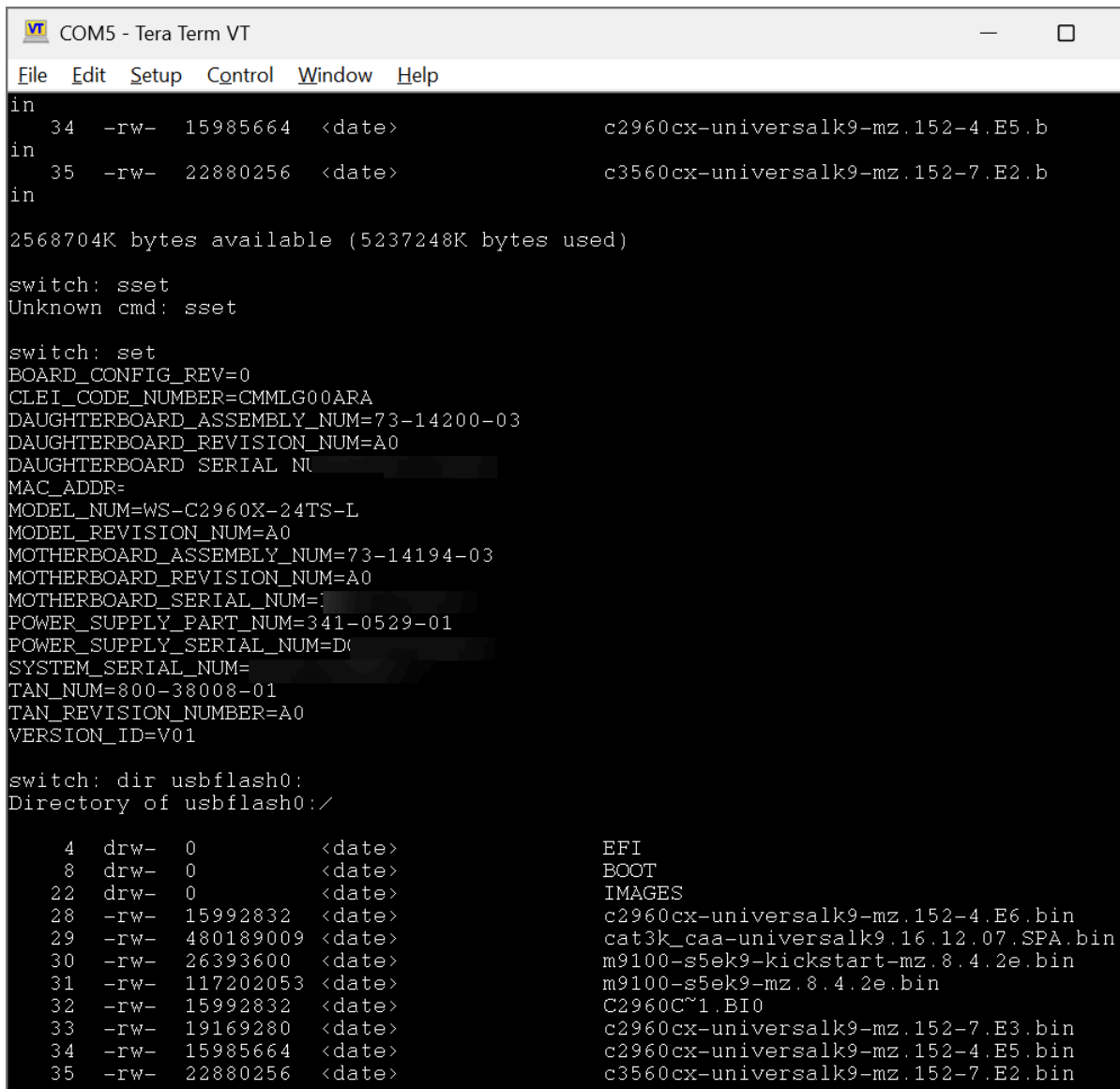
  Proceed with zeroization ?? (yes/[no]): yes
% Proceeding to zeroize image. "Reload" session to remove the loaded image.
Removing usbflash0:c2960x-universalk9-mz.152-7.E11.bin
Oct  9 00:04:58.584: %SYS-7-NV_BLOCK_INIT: Initialized the geometry of nvram
FIPS: Error removing authorization-key.
The configuration is reset and the system will now reboot
Oct  9 00:05:02.991: %SYS-5-RELOAD: Reload requested by Exec. Reload Reason:
Reason unspecified.
CPU rev: B
Image passed digital signature verification
Board rev: 5
```

Figure 4. Redacted original terminal screenshot supplied for this project.

05 | Bootloader environment inspection

Commands observed: `set`; `dir usbflash0`:

The bootloader environment and USB directory were inspected. Serial-number fields in the supplied evidence have been blurred.



```
COM5 - Tera Term VT
File Edit Setup Control Window Help
in
 34 -rw- 15985664 <date> c2960cx-universalk9-mz.152-4.E5.b
in
 35 -rw- 22880256 <date> c3560cx-universalk9-mz.152-7.E2.b
in
2568704K bytes available (5237248K bytes used)

switch: sset
Unknown cmd: sset

switch: set
BOARD_CONFIG_REV=0
CLEI_CODE_NUMBER=CMMLG00ARA
DAUGHTERBOARD_ASSEMBLY_NUM=73-14200-03
DAUGHTERBOARD_REVISION_NUM=A0
DAUGHTERBOARD_SERIAL_NUM=
MAC_ADDR=
MODEL_NUM=WS-C2960X-24TS-L
MODEL_REVISION_NUM=A0
MOTHERBOARD_ASSEMBLY_NUM=73-14194-03
MOTHERBOARD_REVISION_NUM=A0
MOTHERBOARD_SERIAL_NUM=
POWER_SUPPLY_PART_NUM=341-0529-01
POWER_SUPPLY_SERIAL_NUM=D
SYSTEM_SERIAL_NUM=
TAN_NUM=800-38008-01
TAN_REVISION_NUMBER=A0
VERSION_ID=V01

switch: dir usbflash0:
Directory of usbflash0:/

 4 drw- 0 <date> EFI
 8 drw- 0 <date> BOOT
22 drw- 0 <date> IMAGES
28 -rw- 15992832 <date> c2960cx-universalk9-mz.152-4.E6.bin
29 -rw- 480189009 <date> cat3k_caa-universalk9.16.12.07.SPA.bin
30 -rw- 26393600 <date> m9100-s5ek9-kickstart-mz.8.4.2e.bin
31 -rw- 117202053 <date> m9100-s5ek9-mz.8.4.2e.bin
32 -rw- 15992832 <date> C2960C~1.BIO
33 -rw- 19169280 <date> c2960cx-universalk9-mz.152-7.E3.bin
34 -rw- 15985664 <date> c2960cx-universalk9-mz.152-4.E5.bin
35 -rw- 22880256 <date> c3560cx-universalk9-mz.152-7.E2.bin
```

Figure 5. Redacted original terminal screenshot supplied for this project.

06 | USB boot after zeroization

Commands observed: boot usbflash0:c2960x-universalk9-mz.152-7.E11.bin

The approved image was booted again and console output includes image verification activity.

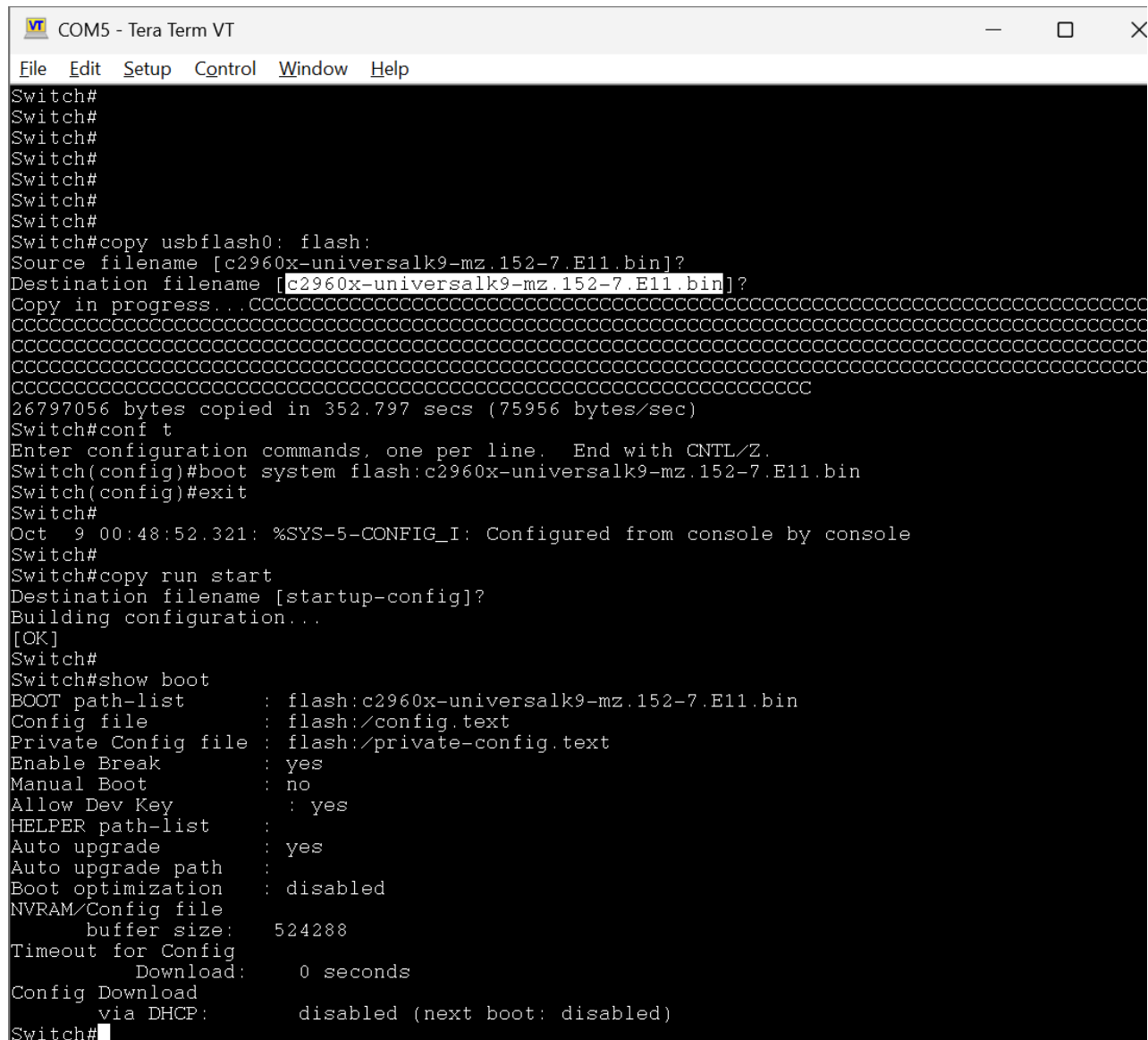
[illegible]

Figure 6. Redacted original terminal screenshot supplied for this project.

07 | Flash recovery and persistent boot configuration

Commands observed: copy usbflash0: flash:; boot system flash:...; copy run start; show boot

Console capture shows the image copied into flash, the boot statement configured, the running configuration copied to startup configuration, and the BOOT path pointing to the internal image.



```
COM5 - Tera Term VT
File Edit Setup Control Window Help
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#copy usbflash0: flash:
Source filename [c2960x-universalk9-mz.152-7.E11.bin]?
Destination filename [c2960x-universalk9-mz.152-7.E11.bin]?
Copy in progress...
26797056 bytes copied in 352.797 secs (75956 bytes/sec)
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#boot system flash:c2960x-universalk9-mz.152-7.E11.bin
Switch(config)#exit
Switch#
Oct  9 00:48:52.321: %SYS-5-CONFIG_I: Configured from console by console
Switch#
Switch#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
Switch#
Switch#show boot
BOOT path-list       : flash:c2960x-universalk9-mz.152-7.E11.bin
Config file          : flash:/config.text
Private Config file  : flash:/private-config.text
Enable Break         : yes
Manual Boot          : no
Allow Dev Key        : yes
HELPER path-list     :
Auto upgrade         : yes
Auto upgrade path    :
Boot optimization    : disabled
NVRAM/Config file    :
  buffer size:       524288
Timeout for Config
  Download:          0 seconds
Config Download
  via DHCP:          disabled (next boot: disabled)
Switch#
```

Figure 7. Redacted original terminal screenshot supplied for this project.

08 | Running configuration inspection

Commands observed: show run

The screenshot displays the start of the running configuration, including switch identification and IOS configuration lines. A partial screenshot is not evidence of all configuration fields.

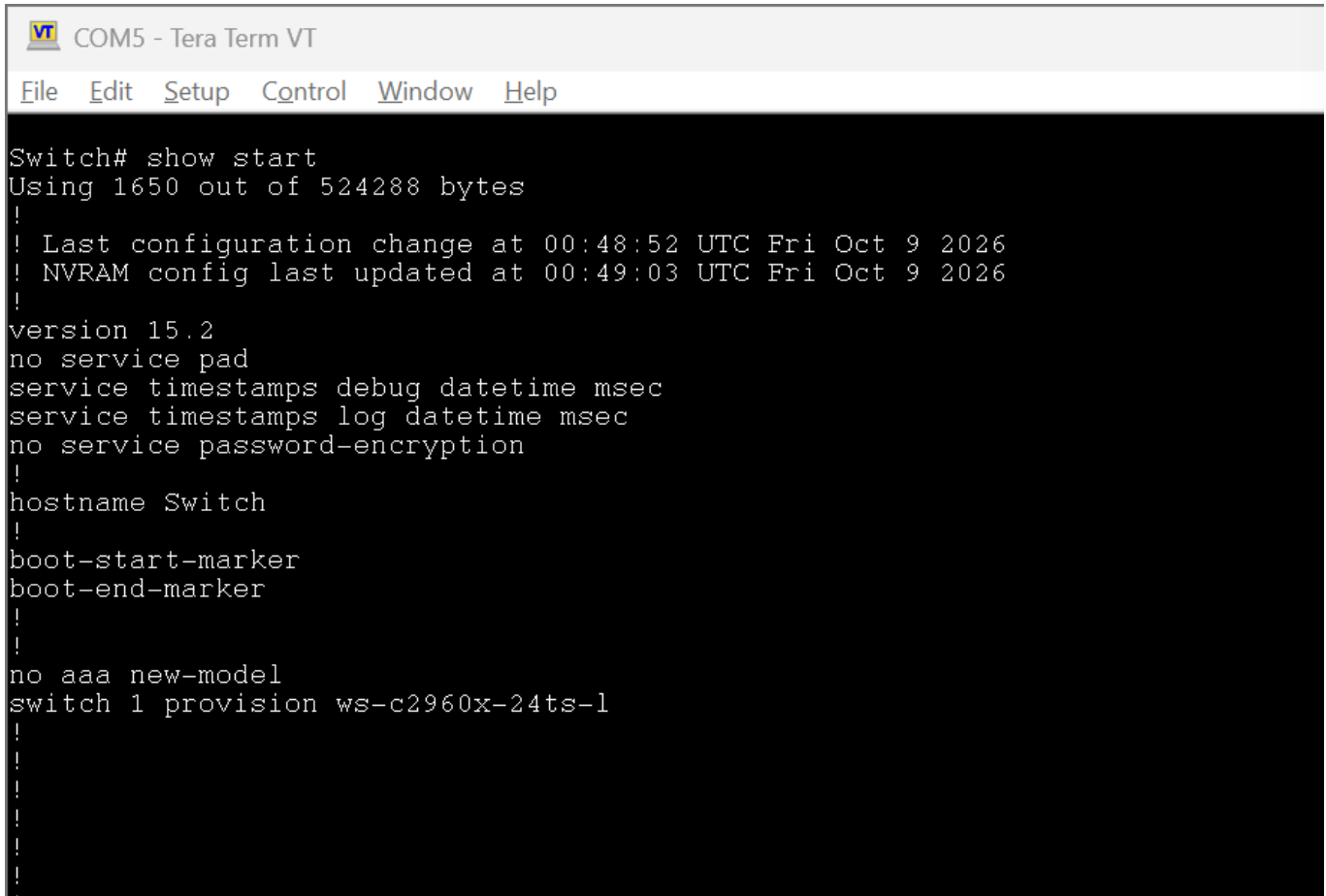
```
VT COM5 - Tera Term VT  
File Edit Setup Control Window Help  
7993294848 bytes total (2513649664 bytes free)  
Switch#  
Switch#  
Switch#  
Switch#SHOW RUN  
Building configuration...  
  
Current configuration : 1650 bytes  
!  
! Last configuration change at 00:48:52 UTC Fri Oct 9 2026  
! NVRAM config last updated at 00:49:03 UTC Fri Oct 9 2026  
!  
version 15.2  
no service pad  
service timestamps debug datetime msec  
service timestamps log datetime msec  
no service password-encryption  
!  
hostname Switch  
!  
boot-start-marker  
boot-end-marker  
!  
!  
no aaa new-model  
switch 1 provision ws-c2960x-24ts-l  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
spanning-tree mode rapid-pvst  
spanning-tree extend system-id
```

Figure 8. Redacted original terminal screenshot supplied for this project.

09 | Startup configuration inspection

Commands observed: `show start`

The screenshot displays the start of the startup configuration. The presence of a saved configuration is intentional in the pictured boot setup and must be reconciled with the required delivery state.



```
COM5 - Tera Term VT
File Edit Setup Control Window Help

Switch# show start
Using 1650 out of 524288 bytes
!
! Last configuration change at 00:48:52 UTC Fri Oct 9 2026
! NVRAM config last updated at 00:49:03 UTC Fri Oct 9 2026
!
version 15.2
no service pad
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname Switch
!
boot-start-marker
boot-end-marker
!
!
no aaa new-model
switch 1 provision ws-c2960x-24ts-l
!
!
!
```

Figure 9. Redacted original terminal screenshot supplied for this project.

10 | VLAN database inspection

Commands observed: `show vlan`

The output lists VLAN 1 and reserved default VLANs. No custom VLAN is visible in the displayed portion.

```
COM5 - Tera Term VT
File Edit Setup Control Window Help

Switch#show vlan

VLAN Name                Status    Ports
-----
1    default                active    Gi1/0/1, Gi1/0/2, Gi1/0/3
                                           Gi1/0/4, Gi1/0/5, Gi1/0/6
                                           Gi1/0/7, Gi1/0/8, Gi1/0/9
                                           Gi1/0/10, Gi1/0/11, Gi1/0/12
                                           Gi1/0/13, Gi1/0/14, Gi1/0/15
                                           Gi1/0/16, Gi1/0/17, Gi1/0/18
                                           Gi1/0/19, Gi1/0/20, Gi1/0/21
                                           Gi1/0/22, Gi1/0/23, Gi1/0/24
                                           Gi1/0/25, Gi1/0/26, Gi1/0/27
                                           Gi1/0/28
1002 fddi-default          act/unsup
1003 token-ring-default    act/unsup
1004 fddinet-default        act/unsup
1005 trnet-default          act/unsup

VLAN Type  SAID      MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
-----
1    enet   100001    1500  -      -      -      -    -          0      0
1002 fddi   101002    1500  -      -      -      -    -          0      0
1003 tr    101003    1500  -      -      -      -    -          0      0

VLAN Type  SAID      MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
-----
1004 fdnet 101004    1500  -      -      -      ieee -          0      0
1005 trnet 101005    1500  -      -      -      ibm  -          0      0

Remote SPAN VLANs
-----

Primary Secondary Type      Ports
-----

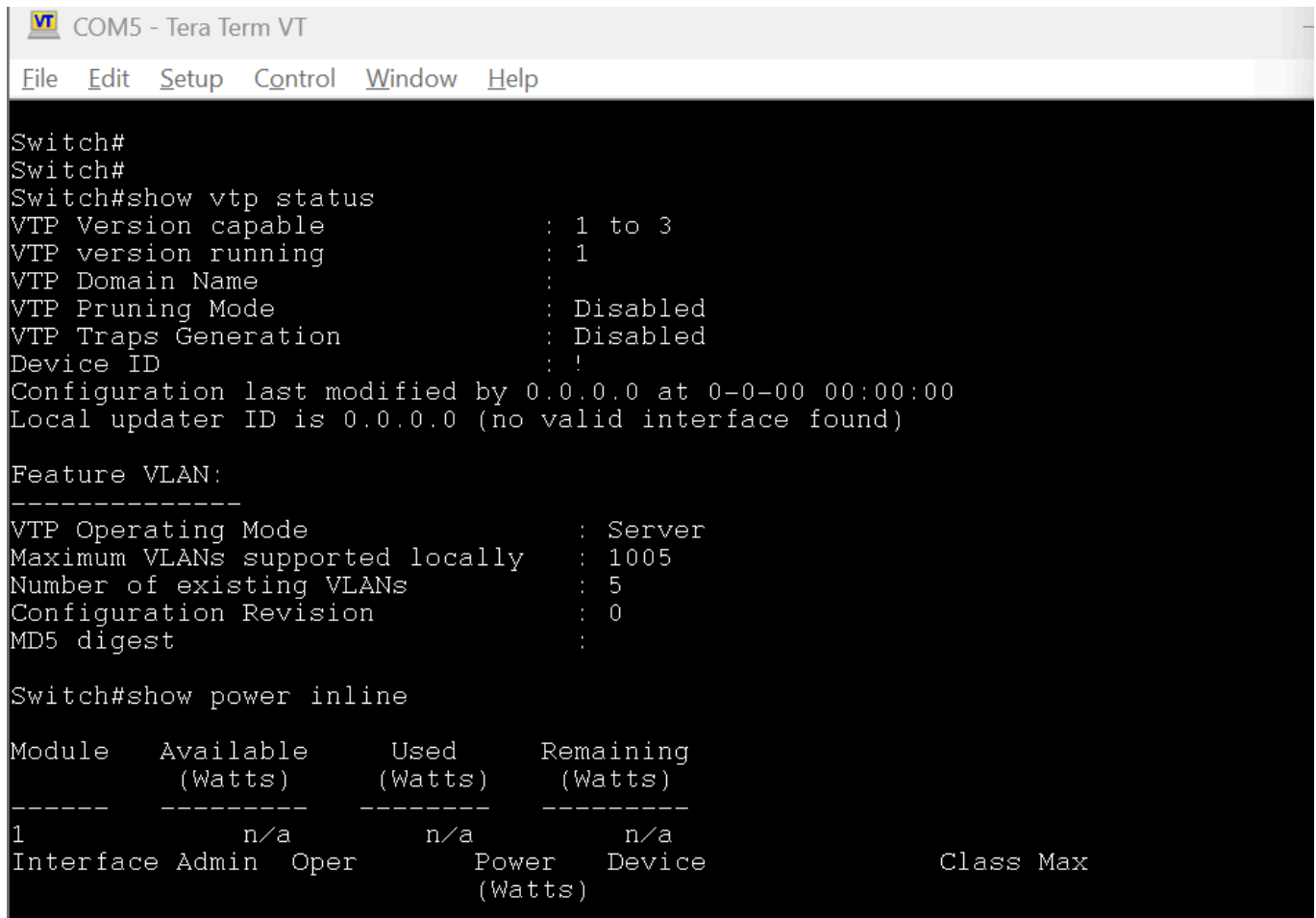
Switch#
Switch#
Switch#show vtp status
VTP Version capable      : 1 to 3
VTP version running      : 1
VTP Domain Name          :
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
```

Figure 10. Redacted original terminal screenshot supplied for this project.

11 | VTP and PoE capability check

Commands observed: `show vtp status`; `show power inline`

The capture shows VTP parameters and PoE output. This 24TS-L model is non-PoE; n/a output is expected, not a failed PoE test.



```
COM5 - Tera Term VT
File Edit Setup Control Window Help

Switch#
Switch#
Switch#show vtp status
VTP Version capable          : 1 to 3
VTP version running         : 1
VTP Domain Name              :
VTP Pruning Mode             : Disabled
VTP Traps Generation         : Disabled
Device ID                    : !
Configuration last modified by 0.0.0.0 at 0-0-00 00:00:00
Local updater ID is 0.0.0.0 (no valid interface found)

Feature VLAN:
-----
VTP Operating Mode           : Server
Maximum VLANs supported locally : 1005
Number of existing VLANs     : 5
Configuration Revision       : 0
MD5 digest                   :

Switch#show power inline

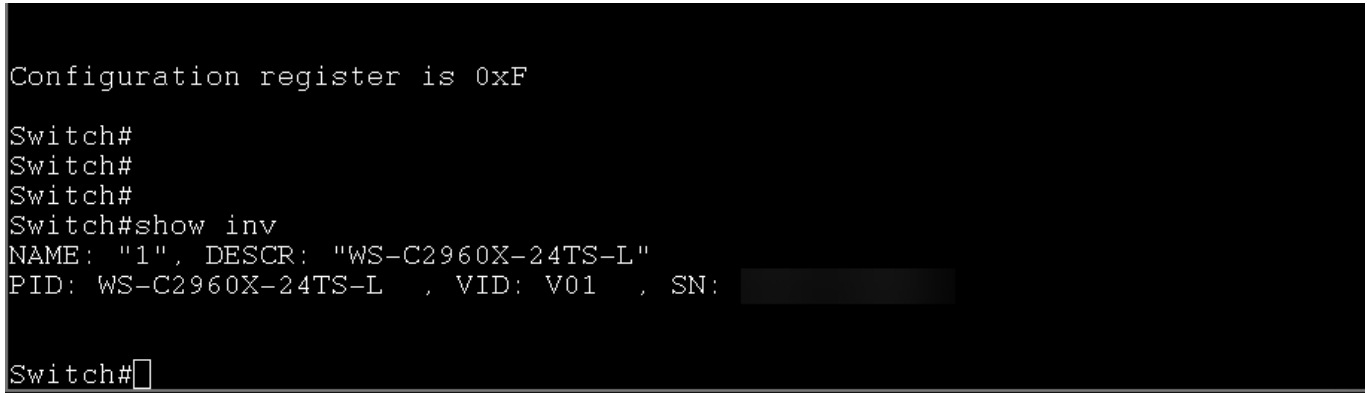
Module      Available      Used      Remaining
  (Watts)    (Watts)    (Watts)
-----
1           n/a           n/a           n/a
Interface Admin Oper      Power Device      Class Max
              (Watts)
-----
```

Figure 11. Redacted original terminal screenshot supplied for this project.

12 | Device identity validation

Commands observed: `show inv`

Inventory output identifies the WS-C2960X-24TS-L model and version. Its serial number has been blurred.

A terminal window with a black background and white text. The text shows a configuration register value, followed by several 'Switch#' prompts, then the command 'show inv'. The output of 'show inv' shows 'NAME: "1", DESCR: "WS-C2960X-24TS-L"', 'PID: WS-C2960X-24TS-L', 'VID: V01', and 'SN:' followed by a blurred area. The prompt 'Switch#' is followed by a cursor.

```
Configuration register is 0xF  
  
Switch#  
Switch#  
Switch#  
Switch#show inv  
NAME: "1", DESCR: "WS-C2960X-24TS-L"  
PID: WS-C2960X-24TS-L , VID: V01 , SN:   
  
Switch#
```

Figure 12. Redacted original terminal screenshot supplied for this project.

Verification and honest status

Validation item	Evidence status
Flash formatting, USB boot, IOS copy	Observed
Secure reset and FIPS zeroization	Commands / console output observed
BOOT path in internal flash	Observed via show boot
VLAN, VTP and configuration inspection	Observed; captured portions only
Hardware identification	Observed; serial blurred
USB-free reboot and port traffic QA	Not evidenced
Checksum compared to trusted reference	Not evidenced
Formal sanitization audit acceptance	Not evidenced

Additional quality-assurance opportunities (not shown in supplied evidence)

With authorization: perform an independent USB-free reload, review environmental/system logs, assess interface counters, and test copper-port traffic with isolated devices. Record pass/fail and limits. The 24TS-L model has no PoE.

Technical interpretation

The screenshots document a concrete recovery and post-operation verification workflow. Boot-path confirmation is visible, but a successful USB-free reboot is not demonstrated. A saved startup configuration appears in the console capture; the required final handoff state must be reconciled with the controlled sanitization process.